

Elastix Parameter File Reference Guide

How to read an Elastix parameter file

Elastix registration is modular. A parameter file selects the registration framework, geometric transform, similarity metric, optimizer, image sampling strategy, multi-resolution image pyramids, interpolation, and final resampling behavior.

Conceptual flow: Registration -> Transform -> Metric -> Optimizer -> Image Sampler -> Image Pyramid -> Interpolator -> Resample Interpolator -> Resampler

1. Registration

The Registration component controls the overall framework that coordinates the other components. For most workflows, a multi-resolution strategy is used: registration starts on a coarse representation and progressively moves toward finer resolution.

Option	Meaning	Practical note
MultiResolutionRegistration	Standard coarse-to-fine registration using one metric configuration.	Main/default-style choice for routine registration.
MultiMetricMultiResolutionRegistration	Multi-resolution registration that can combine multiple metrics.	Useful when similarity and/or penalty terms must be optimized together.
MultiResolutionRegistrationWithFeatures	Multi-resolution framework using multiple feature images.	Advanced/specialized; outside the main workflow of this project.

Key parameters

Parameter	Values / default	Meaning
NumberOfResolutions	Positive integer. Default: 3.	Number of coarse-to-fine resolution levels.
Metric<i>Weight	Numeric. MultiMetric only.	Controls the contribution of metric i when multiple metrics are combined.
UseRelativeWeights	"true" / "false"	Allows relative weighting of multiple metrics instead of only fixed absolute weights.
Metric<i>Use	"true" / "false"	Controls whether a particular metric participates at a given resolution.

Mask-related parameters (fixed/moving masks and mask erosion) are supported by Elastix, but are outside the scope of this guide.

2. Transform

Transform defines the geometric changes that Elastix is allowed to apply to the moving image. Choosing a more flexible transform increases the degrees of freedom and should match the expected physical differences between images.

Transform	What it allows	Typical use
TranslationTransform	Translation only.	Images that differ mainly by position.
EulerTransform	Translation + rotation.	Rigid registration.
SimilarityTransform	Translation + rotation + uniform scaling.	Global size difference while preserving shape.
AffineTransform	Translation + rotation + scaling + shear.	More flexible global geometric differences.
BSplineTransform	Spatially varying local deformation.	Non-rigid anatomical/tissue deformation.

Other specialized transforms exist (for example DTI, stack, deformation-field and combination transforms), but they are outside the main scope of this guide.

Key parameters

Parameter	Values / default	Meaning / recommendation
AutomaticTransformInitialization	"true" / "false". Default: false.	Automatically estimates an initial alignment before optimization.
AutomaticTransformInitializationMethod	e.g. "GeometricalCenter", "CenterOfGravity"	Defines how automatic initialization is calculated.
AutomaticScalesEstimation	"true" / "false".	Automatically estimates optimizer parameter scales; useful when rotation and translation parameters have different numerical effects.
CenterOfRotation / CenterOfRotationPoint	Coordinates	Defines the point around which rotation-based transforms operate.
HowToCombineTransforms	"Compose" / "Add". Default in modern Elastix: "Compose".	Compose applies transforms sequentially; Add combines their effects additively. Compose is the relevant choice for normal staged pipelines.

B-Spline-specific parameters

Parameter	Values / default	Meaning
FinalGridSpacingInVoxels	Numeric spacing; commonly specified per dimension.	Controls spacing between B-spline control points. Smaller spacing gives more local flexibility.
FinalGridSpacingInPhysicalUnits	Physical distance	Same concept expressed in physical units rather than voxels.
BSplineTransformSplineOrder	1, 2 or 3. Default: 3.	Order of the B-spline transform; 3 corresponds to cubic B-splines.

3. Metric

Metric is the objective used to judge alignment quality during optimization. It compares the fixed image with the transformed moving image and returns the value that guides the optimizer.

Metric	Main idea	Typical use
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AdvancedMeanSquares	Penalizes numerical intensity differences.	Same modality with directly comparable intensities.
AdvancedNormalizedCorrelation	Measures correlation of intensity patterns rather than exact equality.	Same modality or approximately linear intensity relationship.
AdvancedMattesMutualInformation	Measures statistical dependence between image intensities.	Very common; useful when intensities do not correspond directly, including multimodal registration.
NormalizedMutualInformation	Normalized mutual-information formulation.	Alternative MI-based similarity measure.
Penalty metrics	Penalize undesirable transformations rather than measuring image similarity alone.	Regularization, especially for non-rigid registration.

Key parameters

Parameter	Meaning
NumberOfHistogramBins	Controls histogram resolution for mutual-information-based metrics. More bins are not automatically better; the value interacts with noise and sampling.
FixedKernelBSplineOrder / MovingKernelBSplineOrder	Advanced settings controlling density estimation for mutual information. Usually left unchanged unless deliberately tuning MI.

For B-spline registration, penalty terms such as bending-energy or rigidity penalties can be combined with a similarity metric to discourage unrealistic deformation. Detailed penalty tuning is considered advanced and is outside the main scope.

4. Optimizer

The optimizer changes the transform parameters iteratively in order to improve the metric. In simple terms: Transform defines what can change, Metric defines what 'better' means, and Optimizer defines how to search for it.

Optimizer	Meaning	Scope
AdaptiveStochasticGradientDescent	Stochastic gradient descent with adaptive step-size behavior.	Primary practical choice; widely used in Elastix.
StandardGradientDescent	Basic gradient-descent optimization.	Useful conceptually / simpler cases.
AdaptiveStochasticLBFGS	Stochastic quasi-Newton optimization using gradient history.	Advanced.
PreconditionedStochasticGradientDescent	Stochastic gradient descent with preconditioning.	Advanced.

Key parameters

Parameter	Meaning
MaximumNumberOfIterations	Maximum optimization iterations per resolution level. Values may be specified separately for each resolution.
AutomaticParameterEstimation	Allows Elastix to estimate important optimizer settings automatically rather than requiring extensive manual tuning.

NewSamplesEveryIteration	When random sampling is used, controls whether a new spatial sample is selected at every iteration.
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Elastix exposes additional optimizer settings for step-size schedules, gradient estimation and preconditioning. These are advanced tuning options and are not expanded in this practical guide.

5. Image Sampler

ImageSampler controls which image locations are used to compute the metric. Sampling can substantially reduce computation time for large 3D volumes.

Sampler	Meaning	Practical note
Full	Uses all available voxels.	Most exhaustive but potentially expensive.
Random	Randomly samples voxels.	Common and efficient general-purpose choice.
RandomCoordinate	Samples random coordinates in image space.	Useful stochastic alternative with interpolation at sampled coordinates.
Grid	Samples locations on a regular grid.	Spatially uniform sampling.

Key parameters

Parameter	Meaning
NumberOfSpatialSamples	Number of sampled locations used for metric estimation. More samples generally improve stability but increase computation.
NewSamplesEveryIteration	If "true", a new random sample is selected for each iteration; mainly relevant to random samplers.

6. Image Pyramid

The fixed and moving image pyramids create the coarse-to-fine image representations used by multi-resolution registration. NumberOfResolutions specifies how many levels exist; the pyramid controls how those levels are generated.

Option	Meaning
Fixed/MovingSmoothingImagePyramid	Applies smoothing at each resolution without necessarily shrinking the image.
Fixed/MovingShrinkingImagePyramid	Downsamples the image.
Fixed/MovingRecursiveImagePyramid	Builds each level recursively from the preceding level.
Fixed/MovingGenericImagePyramid	Flexible pyramid with explicit control over smoothing/rescaling schedules.

Key parameters

Parameter	Meaning
FixedImagePyramidSchedule	Defines the fixed-image scaling schedule across resolution levels and dimensions.
MovingImagePyramidSchedule	Defines the corresponding moving-image schedule.

ImagePyramidSchedule	A shared schedule form used in some parameter sets.
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Example for 3 resolutions in 3D: 4 4 4 / 2 2 2 / 1 1 1. This means registration progresses from a coarse representation toward full resolution.

7. Interpolator

During optimization, transformed coordinates usually fall between voxels. The Interpolator estimates moving-image intensity at those non-grid locations.

Interpolator	Meaning	Practical note
NearestNeighborInterpolator	Uses the nearest voxel value.	Fast but discontinuous; mainly useful for discrete labels rather than intensity optimization.
LinearInterpolator	Linear interpolation between neighboring voxels.	Fast and commonly appropriate.
BSplineInterpolator	Smooth B-spline interpolation.	Common in intensity-based registration.
ReducedDimensionBSplineInterpolator	B-spline interpolation with reduced-dimensional behavior.	Specialized.

Key parameter

Parameter	Meaning
BSplineInterpolationOrder	Order used during registration. Order 1 corresponds to linear interpolation; higher orders are smoother but more computationally expensive.

Important distinction: Interpolator is used during optimization. ResampleInterpolator is used after the transform has been found to generate the final registered image.

8. Resample Interpolator

ResampleInterpolator controls interpolation when producing the final transformed image. Because this step is performed after optimization, it may use a higher-quality interpolation than the one used during iterative registration.

Option	Meaning	Practical note
FinalNearestNeighborInterpolator	Nearest-neighbor interpolation for final output.	Appropriate for masks/labels because it does not create intermediate class values.
FinalLinearInterpolator	Linear interpolation for final output.	Simple and relatively fast.
FinalBSplineInterpolator	B-spline interpolation for final output.	Common choice for continuous intensity images.
FinalReducedDimensionBSplineInterpolator	Reduced-dimensional final B-spline interpolation.	Specialized.

Key parameter

Parameter	Values / recommendation
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FinalBSplineInterpolationOrder	Commonly 3 for continuous intensity images. Order 0 is appropriate for binary/label images when using B-spline final interpolation semantics.
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9. Resampler and Output

The Resampler creates the final registered image on the output grid using the final transform and the selected ResampleInterpolator.

Parameter / option	Meaning
Resampler "DefaultResampler"	Standard Elastix resampler used to generate the result image.
DefaultPixelValue	Value assigned when a transformed output location maps outside the moving image domain.
WriteResultImage	Controls whether the registered image is written to disk ("true"/"false").
ResultImagePixelType	Output pixel type, for example "float" or "short".
ResultImageFormat	Output image format, for example "tif", "mhd", or "nii" depending on build/support and workflow.
CompressResultImage	Controls compression when the selected output format supports it.

10. Practical selection guide

Question	Typical starting choice
Images differ mainly by position?	TranslationTransform
Need rotation but shape/size should remain fixed?	EulerTransform (rigid)
Need a global size/shear correction?	AffineTransform
Need local tissue deformation?	BSplineTransform, preferably after a global alignment stage
Same modality and directly comparable intensities?	AdvancedMeanSquares or AdvancedNormalizedCorrelation
Complex intensity relationship / multimodal images?	AdvancedMattesMutualInformation
Large 3D data?	Random/RandomCoordinate sampling + multi-resolution pyramid
Continuous microscopy intensity output?	FinalBSplineInterpolator
Label or segmentation output?	Nearest-neighbor final interpolation

11. Example staged workflow

A practical registration pipeline can use several parameter files sequentially. Each stage refines the result of the previous stage.

Rigid -> Affine -> B-Spline (optional)

For normal staged registration, transforms are typically combined using `HowToCombineTransforms = Compose`, so the new transform is applied in sequence with the existing transform.

12. Scope notes

Elastix contains many additional specialized components and parameters for masks, DTI, groupwise registration, 2D/3D projection registration, specialized penalties, GPU/OpenCL configurations, transform stacks, and detailed optimizer tuning. They are intentionally not expanded here. This guide is intended to remain readable and useful for the lab's routine 3D microscopy workflow.

Sources used for this draft

Elastix official website and documentation (elastix.dev)

Elastix Model Zoo (elastix.dev/modelzoo.html)

Klein et al., 2010 - elastix: a toolbox for intensity-based medical image registration

Klein et al., 2009 - Adaptive Stochastic Gradient Descent Optimisation for Image Registration

SuperElastix/elastix official project and parameter examples

Draft status: This document is intentionally a first-pass practical reference. Defaults and recommendations should be reviewed against the exact Elastix version and the lab's final parameter files before release.

13. Appendices

Appendix A - Example Elastix Parameter File Example Content

(FixedImageDimension 3)

(MovingImageDimension 3)

(FixedInternalImagePixelType "float")

(MovingInternalImagePixelType "float")

(Registration "MultiResolutionRegistration")

(Transform "AffineTransform")

(AutomaticScalesEstimation "true")

(Metric "AdvancedMeanSquares")

(Optimizer "AdaptiveStochasticGradientDescent")

(FixedImagePyramid "FixedRecursiveImagePyramid")

(MovingImagePyramid "MovingRecursiveImagePyramid")

(NumberOfResolutions 3)

(ImageSampler "Full")

(AutomaticParameterEstimation "true")

(MaximumNumberOfIterations 512)

(Interpolator "BSplineInterpolator")

(BSplineInterpolationOrder 1)

(Resampler "DefaultResampler")

(ResampleInterpolator "FinalBSplineInterpolator")

(FinalBSplineInterpolationOrder 1)

(DefaultPixelValue 0)

(WriteResultImage "true")

(ResultImageFormat "tif")

(ResultImagePixelType "float")